INSTRUCTIONS

- Use Font size 22 to 28 and Font Style Times New Roman (body)
- Do not change the Template: Heading / Title of the slides
- Do not change font style or colours
- Do not copy paste paragraphs: use only Bullet Points
- Maximum slide Limit is 20 only
- As per the number of supervisor allotted adopt slide-2 or slide-3
- Please display the required meaningful Content or images or tables
- Change the footer accordingly using insert tab then click on header & footer tab and edit it
- Save the PPT with Register No, Dept and Review no, example: 23xxxxxxxxx_CSE

Title: Automated expense splitting and sharing

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Under the Guidance
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(DEEMED TO BE UNIVERSITY)

OUTLINE

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Abstract

- The Automated Expense Splitting and Sharing project is a web-based application designed to simplify the management of shared expenses among groups. When people travel, dine together, or share living expenses, calculating individual contributions manually can be difficult and time-consuming.
- This project allows users to enter expenses, select participants, and automatically split the cost among them. The application uses web development technologies and basic data structures to store and process expense information efficiently.

Existing System

- In the existing system, people usually track shared expenses manually using notebooks, calculators, or simple spreadsheets.
- Each person must calculate the amount owed by every participant individually.
- This method is time-consuming and prone to calculation errors.

Proposed System

- Java console-based Automated expense splitting and sharing
- Allows users to add, display, search, sort, and delete expense records
- Uses ArrayList to store expense data and participant details. ArrayList can store expense amounts, user names, and transaction records.
- .Implements Linear Search can be used for searching a participant and a specific expense record.
- Uses Bubble sort to arrange expenses and sorting expenses from highest to lowest.
- Uses Stack to undo the last expense entry
- Uses Queue to process expenses in order and entered processed first(FIFO)
- Uses HashMap to store and calculate each persons share.
- Provides a faster and more organized way to manage shared expenses
- Reduces manual calculation errors and improves expense tracking.
- Helps understand the practical use of Data Structures and Algorithms

System Architecture

- **System architecture** refers to the overall structure of a software application that shows how different components interact with each other to perform the required tasks.
- Input: user enters details(date, number of persons)
- Data storage: stored using arraylist
- Processing: operations like add, sort, delete, search, display
- Algorithms: uses linear search and bubble sort
- Data structures: uses stack, queue, hash map for efficient handling
- Output: displayed on the screen

Project Modules

- . User Registration Module
- Allows new users to create an account.
- Stores user information securely.
- 2. Login Module
- Users can log in with username and password.
- Ensures secure access to the application.
- 3. Expense Entry Module
- Users can add new expenses.
- Enter details like:
  - Amount
  - Description
  - Participants
- 4. Expense Splitting Module
- Automatically divides the total amount among members.
- Shows each person’s share clearly.
- 5. Balance Tracking Module
- Keeps track of who owes money and who should receive money.
- Displays pending balances.
- 6. Settlement Module
- Allows users to mark payments as settled.
- Updates the balance status.

Objectives

- **To develop a web-based application** that helps users easily split
- **To simplify expense calculation** by automatically dividing the total bill amount, including GST or tax, among the involved persons.
- **To provide a user-friendly interface** where users can enter expense details such as category, date, bill amount, and payer information.
- **To maintain a record of previous transactions** so that users can view, edit, or delete past expenses when required.
- **To automatically calculate individual shares** for each participant and display the payable amount clearly.
- **To store expense data securely using Local Storage**, allowing the information to remain available even after refreshing or reopening the web page.
- **To generate a monthly expense summary** that helps users track their total spending over time.
- **To demonstrate the application of HTML, CSS, and JavaScript concepts** such as forms, responsive layouts, DOM manipulation, and data storage.

Co’s Attainments - DSA

1. C0 1: Apply basic data structures and abstract data typed(ADT) to organize and manage data efficiently.
2. CO 2: Implement searching algorithms such as linear search to find specific data in the system
3. CO3: Use sorting algorithms to arrange data in an organized order for easy access and readability
4. C04: Utilize list ADT or linked list concepts for sorting and managing dynamic data.
5. C05: Demonstrate understanding of stack and queue operations for handling data processing tasks.

Co’s Attainments - FWD

CO1 Attainment: Apply internet fundamentals, Html and introductory css styling

C02: Apply Html forms, semantic tags and compressive css layouts

C03: Apply iavascript programming essentials

CO4: Apply javascript interactivity,DOM

C05: Apply advanced web development and deployment

Screenshots for DSA

```html
<!DOCTYPE html>
<html lang="en">
<head>

<meta charset="UTF-8">
<title>Automated Expense Splitting and Sharing</title>

<style>

body{
font-family:Arial;
background:#f4f6f9;
margin:0;
}

header{
background:#2c3e50;
color:white;
text-align:center;
padding:20px;
}

.container{
```

```
margin:auto;
margin-top:20px;
background:white;
padding:20px;
border-radius:8px;
}

input,select,button{
padding:8px;
margin:5px;
}

table{
width:100%;
border-collapse:collapse;
margin-top:20px;
}

th,td{
```

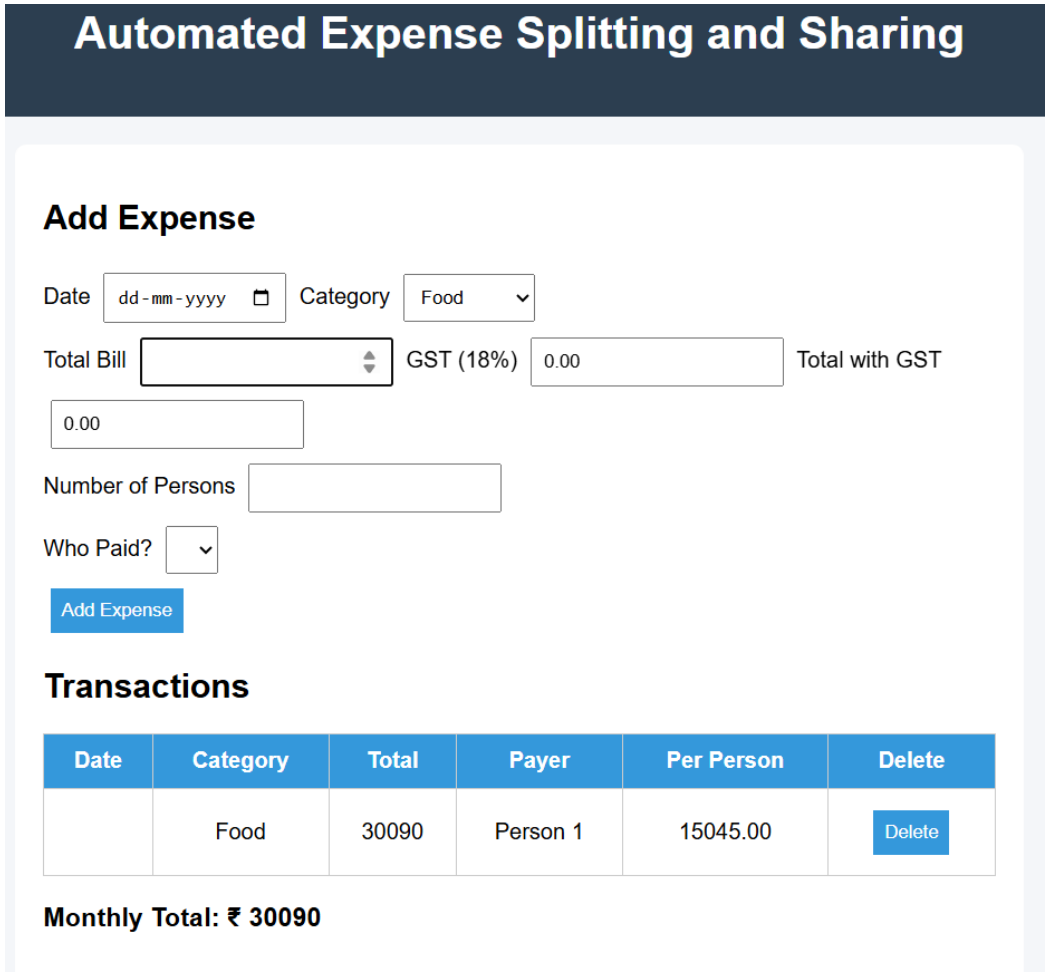
```
button{
background:#3498db;
color:white;
border:none;
cursor:pointer;
}

</style>

</head>

<body>

<header>
<h1>Automated Expense Splitting and Sharing</h1>
</header>

<div class="container">

<h2>Add Expense</h2>
```

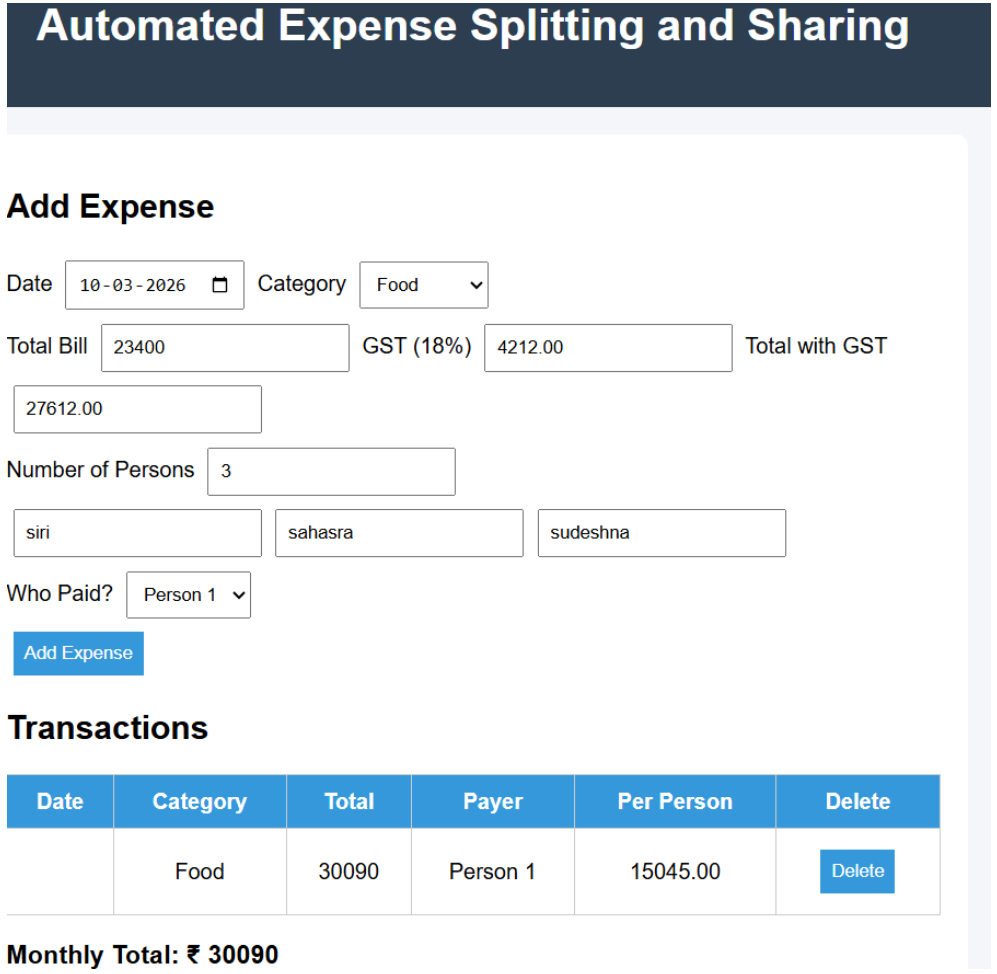
```
<label>Date</label>
<input type="date" id="date">

<label>Category</label>
<select id="category">
<option>Food</option>
<option>Travel</option>
<option>Rent</option>
<option>Shopping</option>
</select>

<br>

<label>Total Bill</label>
<input type="number" id="bill" oninput="calculateGST()">

<label>GST (18%)</label>
<input type="number" id="gst" readonly>
```

```
<script>

let expenses = JSON.parse(localStorage.getItem("expenses")) || []

function calculateGST(){

let bill = Number(document.getElementById("bill").value)

let gst = bill * 0.18

let total = bill + gst

document.getElementById("gst").value = gst.toFixed(2)
document.getElementById("total").value = total.toFixed(2)

}

function createPersons(){
```

```
function del(i){

expenses.splice(i,1)

localStorage.setItem("expenses",JSON.stringify(expenses))

loadTable()

}

loadTable()

</script>

</body>
</html>
```

Screenshots for fwd





Conclusion

- The Automated Expense Splitting and Sharing project simplifies group expense management.
- It helps users split bills accurately and quickly.
- The project reduces manual calculation errors and improves transparency among users.
- It is useful for friends, roommates, travel groups, and event organizers.
- Through this project, we learned web development concepts, expense calculation logic, and user interface design.

Geo- Tag Photos

THANK YOU